R-Based Forecasting Project Using TÜİK Data

Project Instructions

1. Project Overview

In this project, you are expected to develop a small **R-based forecasting project** using a time series data set obtained from the **TÜİK Data Portal**.

You will select one unique TÜİK time series data set, access the data directly through R using the `tuikr` package, use the most recently available TÜİK data, apply alternative forecasting methods, compare the forecasting performance of these methods, select the superior method, and forecast the next available period.

The final project must be published in your own **public GitHub repository**.

The repository must be complete, reproducible, clearly documented, and ready to run.

2. Data Access Requirement

Students must first identify a suitable time-series dataset from the TÜİK Data Portal and then access the same data directly in R using the `tuikr` package.

Package source: https://github.com/emraher/tuikr

Manual data download, copy-paste data entry, manually prepared Excel files, manually prepared CSV files, manually created data sets, or any other non-reproducible data access method are **not allowed**. Any filtering, formatting, time-period arrangement, or variable selection required for forecasting must be performed only within the R notebook through reproducible R code.

Students must create the TÜİK data connection within the R notebook. The project must clearly document the selected TÜİK theme, table name, dataflow ID, selected variable, data frequency, latest available observation, forecast target period, and data access date.

3. Data Set Selection Rules

Each student must select **a different TÜİK time-series dataset**.

The selected data set must satisfy the following requirements:

1. It must be published by TÜİK.
2. It must be accessible through the `tuikr` package.
3. It must be a time-series dataset.
4. The selected variable must be observed repeatedly over ordered time periods.
5. The data must have a clearly defined publication frequency, such as monthly, quarterly, annual, or another regular TÜİK publication period.
6. **The most recently available version** of the selected TÜİK data must be used.
7. The forecast must be produced for **the next period after the latest available observation.**
8. The selected series must contain enough historical observations to apply and compare forecasting methods.
9. The selected data set or selected variable must not be the same as another student's selection.

Before starting the project, each student must register the selected data set in the shared Google Sheet.

4. Forecasting Target

The project aims to forecast the **next unpublished period** using the latest available TÜİK data.

The forecast period must be determined according to the frequency and the latest observation of the selected time series.

Examples:

Data Frequency	Latest Available Observation	Required Forecast Period
Annual	2025	2026
Quarterly	2026 Q1	2026 Q2
Monthly	March 2026	April 2026

Students must explicitly report:

1. the latest available observation in the TÜİK data,
2. the next period to be forecasted,
3. the forecasted value for that next period.

5. Shared Google Sheet Requirement

A shared Google Sheet will be used to prevent duplicate data set selection. Each student must enter the following information into the shared sheet:

Field	Required Information
Student Name	Full name of the student
Student Number	Student ID
TÜİK Data Set Name	Exact name of the selected TÜİK data set
TÜİK Theme / Category	Theme or statistical category
TÜİK Table Name	Exact table name identified through `tuikr`
`tuikr` Dataflow ID	Dataflow ID used to access the data
Selected Variable	Variable to be forecasted
Data Frequency	Monthly, quarterly, annual, etc.
Time Coverage	First and last available period
Latest Available Observation	The most recent period available in the TÜİK data
Forecast Target Period	Next period to be forecasted

No two students may work with the same selected variable from the same TÜİK table.

6. Use of TÜİK Data in R

After importing the selected TÜİK data through `tuikr`, students must use the imported data directly within R for the forecasting analysis. If any adjustment is necessary for forecasting, it must be done only within the R notebook through reproducible R code. Acceptable R-based adjustments include:

1. selecting the variable to be forecasted,
2. identifying the time or period variable,
3. confirming the data frequency,
4. arranging observations chronologically,
5. filtering the relevant series if the imported TÜİK table contains multiple variables or categories,
6. checking whether the selected variable is numeric,
7. checking missing values, missing periods, duplicate periods, or irregular observations,
8. transforming the imported TÜİK output into a structure that can be analyzed as a time series in R.

Students must clearly explain in the notebook what adjustments were made in R and why they were necessary.

7. Required Forecasting Methods

Students must apply the following quantitative forecasting methods where applicable.

If a method is not applicable due to the structure of the selected data, the student must clearly explain why. A method must not be omitted silently.

7.1 Naïve Forecasting

Required outputs:

- Forecast values
- Forecast errors
- Accuracy measures
- Actual vs forecast plot
- Short interpretation

7.2 Moving Average

Required outputs:

- Selected moving average window
- Justification of the selected window size
- Forecast values
- Forecast errors
- Accuracy measures
- Actual vs forecast plot
- Short interpretation

7.3 Weighted Moving Average

Required outputs:

- Selected weights
- Explanation of the weighting logic
- Forecast values
- Forecast errors
- Accuracy measures
- Actual vs forecast plot
- Short interpretation

The weights should normally give more importance to recent observations, unless the student provides a clear justification for a different weighting logic.

7.4 Exponential Smoothing

Required outputs:

- Tested smoothing parameter value or values
- Forecast values
- Forecast errors
- Accuracy measures
- Actual vs forecast plot
- Short interpretation

Students should explain why the selected smoothing parameter is reasonable for the data.

7.5 Trend-Adjusted Exponential Smoothing

Students must apply trend-adjusted exponential smoothing if the series shows a meaningful trend. Required outputs:

- Trend assessment
- Smoothing parameter for the level component
- Smoothing parameter for the trend component
- Forecast values
- Forecast errors
- Accuracy measures
- Actual vs forecast plot
- Short interpretation

If the data do not display a meaningful trend, the student must explain why this method is not suitable.

7.6 Linear Trend Projection

Required outputs:

- Estimated trend equation
- Interpretation of the intercept
- Interpretation of the slope
- Forecast values
- Forecast errors
- Accuracy measures
- Actual vs forecast plot
- Forecast for the next period
- Short interpretation

7.7 Seasonal Indices

This method is generally applicable to monthly, quarterly, or other recurring seasonal data.

Required outputs:

- Number of seasons
- Seasonal indices
- Interpretation of high-season and low-season periods

- Deseasonalized values, if applicable
- Seasonally adjusted forecast, if applicable
- Short interpretation

If the selected data are annual or do not have a recurring seasonal structure, the student must explain why seasonal indices are not applicable.

7.8 Additive Decomposition

Required outputs:

- Trend component
- Seasonal component
- Random or residual component
- Forecast values, if applicable
- Accuracy measures
- Actual vs forecast plot
- Short interpretation

If additive decomposition is not suitable, the student must explain why.

7.9 Multiplicative Decomposition

Required outputs:

- Trend component
- Seasonal component
- Random or residual component
- Forecast values, if applicable
- Accuracy measures
- Actual vs forecast plot
- Short interpretation

If multiplicative decomposition is not suitable, the student must explain why.

7.10 Regression with Trend and Seasonal Dummy Variables

Students must estimate a regression-based forecasting model using time and seasonal dummy variables if the selected data frequency supports seasonal modeling. Required outputs:

- Time trend variable
- Seasonal dummy variables
- Estimated regression equation
- Coefficients table
- Interpretation of the trend coefficient
- Interpretation of seasonal dummy variables
- Forecast values
- Forecast errors

- Accuracy measures
- Actual vs forecast plot
- Short interpretation

If the data are annual or do not contain a seasonal structure, the student must explain why seasonal dummy variables are not applicable.

8. Required Forecast Accuracy Measures

Students must compare the candidate forecasting methods using the following accuracy and monitoring measures:

1. Forecast Error
2. Bias / Mean Error
3. MAD – Mean Absolute Deviation
4. MSE – Mean Squared Error
5. MAPE – Mean Absolute Percent Error
6. RSFE – Running Sum of Forecast Errors
7. Tracking Signal

The comparison must be presented in a clear table. Minimum expected comparison table:

Method	**Bias**	**MAD**	**MSE**	**MAPE**	**RSFE**	**Tracking Signal**	**Next-Period Forecast**
Naïve Forecasting							
Moving Average							
Weighted Moving Average							
Exponential Smoothing							
Trend-Adjusted Exponential Smoothing							
Linear Trend Projection							
Seasonal Indices							
Additive Decomposition							
Multiplicative Decomposition							
Regression with Trend and Seasonal Dummies							

If a method is not applicable, the table should indicate this clearly, and the reason must be explained in the notebook and README.

9. Superior Method Selection

Students must select one superior forecasting method. The superior method must be selected based on both:

1. quantitative forecast accuracy results,
2. suitability of the method to the structure of the selected time series.

Students must not select a method only because it has the lowest error value. The method must also be appropriate for the observed structure of the data. The justification must consider:

- whether the series has trend,
- whether the series has seasonality,
- whether the method captures the observed pattern,
- whether forecast errors are acceptable,
- whether the tracking signal indicates systematic bias,
- whether the actual vs forecast plots support the model choice,
- whether the selected method is reasonable for the data frequency and number of observations.

The final explanation must clearly answer: Why is this method superior for this specific TÜİK time series?

10. Final Forecast Requirement

After selecting the superior method, students must use that method to forecast the next available period. The forecast must be based on the **latest available TÜİK observation** at the time of analysis. The final forecast must be clearly reported in both the notebook and the README file. The final result must include:

1. selected superior method,
2. date of data access,
3. latest available TÜİK observation,
4. forecast target period,
5. forecasted value,
6. short interpretation of the forecast.

11. Plot Requirements

Students must include plots for all candidate methods. Each plot must compare actual and forecast values. At minimum, the project must include:

1. Actual time series plot
2. Actual vs Naïve Forecast plot
3. Actual vs Moving Average plot
4. Actual vs Weighted Moving Average plot

5. Actual vs Exponential Smoothing plot
6. Actual vs Trend-Adjusted Exponential Smoothing plot, if applicable
7. Actual vs Linear Trend Projection plot
8. Actual vs Seasonal Indices forecast plot, if applicable
9. Actual vs Additive Decomposition plot, if applicable
10. Actual vs Multiplicative Decomposition plot, if applicable
11. Actual vs Regression with Trend and Seasonal Dummies plot, if applicable
12. Final plot showing the superior method and the next-period forecast

Each plot must include:

- clear title,
- axis labels,
- legend,
- readable time scale,
- clear distinction between actual and forecast values.

Plots must be saved in the repository under the `outputs/figures/` folder.

12. Notebook Requirement

The main project file must be prepared in notebook format.

Required main file format: .Rmd

Required rendered output format: .html

The notebook must include:

1. Project title
2. Student information
3. TÜİK data source information
4. Data access process through `tuikr`
5. Use of TÜİK data in R
6. Exploratory time series analysis
7. Application of required forecasting methods
8. Forecast accuracy calculations
9. Model comparison table
10. Actual vs forecast plots
11. Superior method selection
12. Final next-period forecast
13. Interpretation of results
14. Limitations
15. Reproducibility notes

<u>The notebook must include code comments.</u> Comments should explain the purpose of the main steps. Unexplained and uncommented code will be considered incomplete.

13. GitHub Repository Requirement

Students must publish their project in their own **public GitHub repository**. The GitHub repository link will be submitted as the final project submission. The repository must be:

1. Public
2. Complete
3. Organized
4. Reproducible
5. Ready to run
6. Clearly documented
7. Supported with tables, figures, and interpretations

14. Expected Repository Structure

Each student repository must follow the structure below:

```
tuik-forecasting-project/
│
├── README.md
├── forecasting_project.Rmd
├── forecasting_project.html
├── outputs/
│   ├── tables/
│   │   ├── accuracy_comparison.csv
│   │   └── final_forecast.csv
│   └── figures/
│       ├── actual_series_plot.png
│       ├── naive_forecast_plot.png
│       ├── moving_average_plot.png
│       ├── weighted_moving_average_plot.png
│       ├── exponential_smoothing_plot.png
│       ├── trend_adjusted_smoothing_plot.png
│       ├── trend_projection_plot.png
│       ├── seasonal_indices_plot.png
│       ├── additive_decomposition_plot.png
│       ├── multiplicative_decomposition_plot.png
│       ├── regression_seasonal_dummy_plot.png
│       └── superior_method_plot.png
├── R/
│   ├── data_import.R
│   ├── forecasting_methods.R
│   ├── accuracy_measures.R
│   └── plots.R
├── renv.lock
└── .gitignore
```

Additional files may be included if needed, but the required files and folders must be present. There must be **no separate data folder** containing manually downloaded, manually edited, or newly created data files.

15. Explanation of Required Repository Components

`README.md`

The README file is the main documentation file of the project.

It must explain:

- what the project is about,
- which TÜİK data set was used,
- how the data were accessed through `tuikr`,
- which variable was forecasted,
- what the latest available observation is,
- what the forecast target period is,
- which forecasting methods were applied,
- how the methods were compared,
- which method was selected as superior,
- what the final forecast is,
- how the project can be reproduced.

`forecasting_project.Rmd`

This is the main R notebook file.

It must include the full analysis, including code, outputs, plots, and written interpretation.

The notebook must show that the data were accessed directly from TÜİK through `tuikr`.

`forecasting_project.html`

This is the rendered version of the R Markdown notebook.

It must show the completed analysis, including tables, plots, model comparison, interpretations, and final forecast.

`outputs/tables/accuracy_comparison.csv`

This file must include the model comparison table with all applicable candidate forecasting methods and required accuracy measures.

`outputs/tables/final_forecast.csv`

This file must include the final forecast result.

It should contain:

- selected superior method,
- date of data access,
- latest available TÜİK observation,
- forecast target period,
- forecasted value,
- key accuracy measure values.

`outputs/figures/`

This folder must include actual vs forecast plots for all candidate methods.

The superior method must also have a final plot showing the actual series, fitted or forecast values, and the next-period forecast.

`R/` Folder

The `R/` folder may include supporting R scripts. The main analysis must still be presented in the notebook, but students may use separate scripts to organize reusable functions or workflows.

Expected script purposes:

File	Purpose
`data_import.R`	Data access process using `tuikr`
`forecasting_methods.R`	Forecasting method workflows
`accuracy_measures.R`	Forecast accuracy calculations
`plots.R`	Plot generation

`renv.lock`

The repository should include an `renv.lock` file so that another user can restore the same R package environment.

`.gitignore`

The repository must include a `.gitignore` file to avoid uploading unnecessary local files.

16. Required README.md Structure

The README file must follow the structure below.

TÜİK Forecasting Project

1. Project Overview

Briefly explain the purpose of the project.

2. Data Source and TÜİK Connection

Explain that the data were accessed directly from the TÜİK Data Portal using the `tuikr` R package.

Include:

- TÜİK data set name:
- TÜİK theme/category:
- TÜİK table name:
- TÜİK dataflow ID:
- Selected variable:
- Data frequency:
- Time coverage:
- Latest available observation:
- Forecast target period:
- Date of data access:
- R package used for data access: `tuikr`
- Package source: https://github.com/emraher/tuikr

3. Research Objective

Explain what variable is being forecasted and why this variable is meaningful.

4. Use of TÜİK Data in R

Explain how the TÜİK data imported through `tuikr` were used in R for forecasting.

Clarify that no manually prepared, manually edited, or newly created data set was used.

Briefly describe:

- selected variable,
- time or period variable,
- data frequency,
- latest available observation,
- forecast target period,
- any R-based filtering or formatting required for the analysis.

5. Exploratory Time Series Analysis

Discuss:

- trend,
- seasonality,
- cyclical movements if visible,
- random variation,
- missing values,
- outliers.

6. Forecasting Methods Applied

List the methods applied in the project:

- Naïve Forecasting
- Moving Average
- Weighted Moving Average
- Exponential Smoothing
- Trend-Adjusted Exponential Smoothing
- Linear Trend Projection
- Seasonal Indices
- Additive Decomposition
- Multiplicative Decomposition
- Regression with Trend and Seasonal Dummy Variables

For each method, briefly explain whether it is applicable to the selected data.

7. Forecast Accuracy Comparison

Present the model comparison table including:

- Bias / Mean Error
- MAD
- MSE
- MAPE
- RSFE
- Tracking Signal

8. Selection of the Superior Method

Explain which method was selected as superior and justify the selection using both numerical accuracy and suitability to the data.

9. Final Next-Period Forecast

Report:

- selected superior method,
- date of data access,
- latest available TÜİK observation,

- forecast target period,
- forecasted value.

10. Interpretation of Results

Interpret the forecast result in plain language.

11. Limitations

Discuss the limitations of the analysis.

Possible limitations may include:

- short time series,
- structural breaks,
- economic shocks,
- data revisions,
- weak seasonality,
- high volatility,
- limited explanatory variables.

12. Reproducibility

Explain how another user can run the project.

13. Repository Structure

Briefly describe the files and folders in the repository.

14. Author

Include:

- student name,
- student number,
- course name.

17. Interpretation Requirement

The project must not consist only of code and outputs. Students must interpret their results. At minimum, the student must discuss:

1. What the selected TÜİK variable represents.
2. Why this variable is suitable for forecasting.
3. Whether the series has trend.
4. Whether the series has seasonality.
5. Which methods performed better.

6. Which methods performed worse.
7. Why the superior method was selected.
8. What the next-period forecast means.
9. What limitations should be considered.

18. Reproducibility Requirement

Another user should be able to open the GitHub repository and understand how to run the project. The project must therefore include:

1. Public GitHub access
2. Clear README file
3. R Markdown notebook
4. Rendered HTML output
5. Required folders
6. TÜİK data access through `tuikr`
7. Output tables
8. Output plots
9. Package and environment information
10. Explanation of how to reproduce the analysis

A project that cannot be run, understood, or reproduced from the repository will be considered incomplete.

19. Minimum Submission Requirements

The submitted project must include the following:

- Public GitHub repository
- Unique TÜİK time series data set
- Data registered in Google Sheet
- Data accessed directly through `tuikr`
- No manually created, edited, uploaded, or saved data set
- Latest available TÜİK data used
- Forecast for the next period after the latest TÜİK observation
- R Markdown notebook
- Rendered HTML output
- Code comments
- Required forecasting methods applied or justified as not applicable
- Accuracy comparison table
- Actual vs forecast plots
- Superior method selected
- Result interpretation
- README.md
- Repository structure
- Reproducibility information

20. Warning on the Use of Artificial Intelligence Tools

Students may use artificial intelligence tools to support learning, coding, debugging, writing, documentation, or interpretation. However, the submitted project must be personally reviewed, technically tested, and academically owned by the student.

AI use does not remove the student's responsibility for the accuracy, originality, consistency, reproducibility, and completeness of the submitted work.

Projects that contain clear signs of unreviewed AI-generated, copied, or template-based submission may not be accepted for evaluation. In such cases, the assignment may receive no grade, regardless of whether some parts of the notebook, README, or code appear technically correct.

The following situations are considered unacceptable.

20.1 Unremoved AI or Template Placeholders

The submission must not contain placeholders or template remnants such as:

- Author: [Your Name]
- your-username
- YOUR_USERNAME
- student_name
- example_project
- your_project_name
- your_dataflow_id
- selected_dataflow_id_here
- your_tuik_dataset
- your_variable
- TODO
- [cite: 1]
- insert plot here
- replace this later

Leaving such expressions in the notebook, README, comments, file names, repository description, or outputs indicates that the project was not properly reviewed before submission.

These are not minor formatting mistakes. They are signs of an unfinalized and academically unreliable submission.

20.2 AI-Generated Meta-Comments Left in the Project

The README, notebook, code comments, or documentation must not include phrases that appear to come from an AI assistant or a template conversation, such as:

- as your instructor requested...

- according to the prompt...
- according to the slides you shared...
- if your teacher asks for real data...
- you can replace this later...
- this is an example project...
- here is a sample README...
- this code should work...

The submitted text must be written as a final academic project, not as an AI draft, instruction note, or unfinished template.

20.3 Wrong Author Name, Wrong Repository, or Another Student's Work

The GitHub repository must clearly belong to the submitting student and must match the student's own project.

The following cases will be treated as serious academic integrity problems:

- another person's name appears as the project author,
- the submitted GitHub link belongs to another student,
- the repository owner, README, notebook, or code contains a different student's information,
- the project title in the submission form does not match the GitHub repository,
- the student submits a copied or slightly modified version of another student's repository,
- the selected TÜİK data set, variable, code structure, README text, and results are nearly identical to another student's work without justification.

If the submitted link does not lead to the student's own work, the assignment may not be evaluated.

20.4 TÜİK Data Not Accessed Correctly Through `tuikr`

This project requires students to access TÜİK data directly from R using the `tuikr` package. The following cases are unacceptable:

- manually downloading TÜİK data as Excel, CSV, or another file,
- copying and pasting TÜİK data into R,
- manually creating a separate data set,
- using a data file prepared outside R as the basis of the analysis,
- using a data source other than TÜİK,
- using a data set that is not accessed through `tuikr`,
- claiming that `tuikr` was used while the notebook does not show a reproducible TÜİK data access process,
- using an example data set instead of the selected TÜİK time series.

If the selected TÜİK data are not accessed correctly and reproducibly through `tuikr`, the project may not be evaluated.

20.5 Creating a New Data Set Instead of Using the TÜİK Data

Students must not create, save, upload, or submit a new data set as the basis of the analysis. Any filtering, formatting, variable selection, period ordering, or time series structuring required for forecasting must be performed only within the R notebook through reproducible R code.

The following cases are unacceptable:

- preparing a separate Excel or CSV file and analyzing that file,
- manually editing the TÜİK data outside R,
- uploading a manually cleaned data file to the repository,
- replacing the TÜİK data with a simplified artificial data set,
- adding invented observations to extend the time series,
- changing published values manually,
- deleting observations without explanation,
- generating synthetic values and treating them as TÜİK data.

The purpose of the project is not to create a new data set. The purpose is to use the most recent TÜİK time series data directly accessed through R.

20.6 Not Using the Latest Available TÜİK Data

Students must use the most recently available version of the selected TÜİK data at the time of analysis.

The forecast must be produced for the next period after the latest available observation.

The following cases are unacceptable:

- using an outdated version of the TÜİK series without explanation,
- stopping the time series before the latest available period,
- forecasting a period that is not the immediate next period,
- using annual data ending in 2024 when 2025 is already available,
- using quarterly data ending in 2025 Q4 but forecasting 2026 Q3 instead of 2026 Q1,
- using monthly data ending in March 2026 but forecasting May 2026 instead of April 2026,
- failing to report the latest available observation and forecast target period.

The README and notebook must clearly state:

- latest available TÜİK observation,
- forecast target period,
- date of data access.

20.7 Selected Data Are Not a Time Series

The selected TÜİK data must be a time series. The selected variable must be observed repeatedly over ordered time periods, such as years, quarters, months, or another regular publication period.

The following cases are unacceptable:

- selecting a cross-sectional data set without a time dimension,
- selecting a table that contains only one period,
- selecting a categorical summary that cannot be forecasted,
- selecting a variable that is not numeric or cannot be meaningfully forecasted,
- failing to identify the time or period variable,
- failing to define the data frequency.

If the selected data are not suitable for time series forecasting, the project may not be evaluated.

20.8 README and Repository Structure Do Not Match

The README must accurately describe the actual repository. It is not acceptable to write an ideal repository structure in the README while the actual files are missing or located elsewhere. For example, if the README says the project includes:

- forecasting_project.Rmd
- forecasting_project.html
- outputs/tables/accuracy_comparison.csv
- outputs/tables/final_forecast.csv
- outputs/figures/
- R/

then these files and folders must actually exist in the repository. A README generated from a generic template but not aligned with the real project will be considered a serious documentation problem.

20.9 Code and Running Instructions Are Inconsistent

The code must be executable using the instructions provided in the README. The following problems are unacceptable:

- README says to run a file that does not exist,
- README refers to Python commands instead of R commands,
- README says the project uses `tuikr`, but the notebook does not use `tuikr`,
- notebook depends on local files that are not in the repository,
- code uses absolute file paths from the student's computer,
- output folders are mentioned but not created or used,
- README gives a GitHub clone command containing `your-username` or an unrelated repository name,

- the rendered HTML file does not correspond to the submitted R Markdown notebook.

Students must test their code before submission.

20.10 Code Does Not Run

The R Markdown notebook must run from beginning to end. The following issues indicate that the code was not properly reviewed:

- missing packages without explanation,
- undefined variables,
- broken function calls,
- syntax errors,
- incomplete code chunks,
- repeated code blocks pasted together,
- object names that change inconsistently across the notebook,
- code that produces no output despite the README claiming otherwise,
- forecasting methods listed in the README but not actually implemented,
- plots listed in the README but not generated,
- accuracy tables claimed in the README but not produced.

If the code does not run, the project may not be evaluated.

20.11 Forecasting Methods Are Missing or Only Superficially Applied

Students are expected to apply the required forecasting methods where applicable. The following methods must be considered:

- Naïve Forecasting
- Moving Average
- Weighted Moving Average
- Exponential Smoothing
- Trend-Adjusted Exponential Smoothing
- Linear Trend Projection
- Seasonal Indices
- Additive Decomposition
- Multiplicative Decomposition
- Regression with Trend and Seasonal Dummy Variables

The following cases are unacceptable:

- applying only one method and calling it a complete forecasting study,
- listing methods in the README without implementing them,
- omitting methods without explanation,
- claiming a method is “not applicable” without technical justification,
- applying seasonal methods to annual data without explanation,
- applying seasonal dummy variables without identifying the season structure,
- applying decomposition without sufficient seasonal observations,

- using functions automatically without interpreting the results,
- selecting the superior method without comparing alternatives.

A method may be excluded only if the student clearly explains why it is not applicable to the selected TÜİK time series.

20.12 Accuracy Comparison Is Missing or Incorrect

Students must compare candidate forecasting methods using the required accuracy and monitoring measures:

- Forecast Error
- Bias / Mean Error
- MAD
- MSE
- MAPE
- RSFE
- Tracking Signal

The following cases are unacceptable:

- no accuracy comparison table,
- comparing methods only visually without error measures,
- reporting only one metric without justification,
- using inconsistent actual and forecast periods,
- calculating errors using mismatched time periods,
- comparing models on different subsets of the data without explanation,
- reporting accuracy values that cannot be reproduced from the code,
- selecting the superior method without reference to the accuracy results.

The comparison table must be reproducible from the submitted notebook.

20.13 Actual vs Forecast Plots Are Missing or Misleading

The project must include actual vs forecast plots for all candidate methods where applicable. The following cases are unacceptable:

- no actual vs forecast plots,
- only plotting the actual time series,
- only plotting the final selected method,
- using unclear axes or missing legends,
- plotting forecast values against the wrong time periods,
- showing plots in the README that are not generated by the notebook,
- using generic or copied plots unrelated to the selected TÜİK data,
- presenting plots that cannot be reproduced from the code.

Each plot must clearly distinguish actual values and forecast values.

20.14 Strong README but Weak, Missing, or Unrelated Code

A polished README is not sufficient if the implementation is incomplete, unrelated, or non-reproducible. The following inconsistencies are unacceptable:

- README describes several forecasting methods, but the notebook applies only one,
- README claims that the latest TÜİK data were used, but the notebook uses an older subset,
- README reports a final forecast that cannot be reproduced from the code,
- README describes actual vs forecast plots, but the repository does not contain them,
- README claims `tuikr` data access, but the notebook reads a local file,
- README includes interpretation that is not based on the actual output,
- README describes a different variable, frequency, or dataflow ID than the notebook.

The project will be evaluated as a complete system: README, notebook, code, outputs, visualizations, and interpretation must be consistent.

20.15 Empty, Generic, or Superficial README

The README must not be empty or filled with generic AI-style text. A valid README should explain the actual project. It must include:

- project overview,
- TÜİK data set name,
- TÜİK theme or category,
- TÜİK table name,
- `tuikr` dataflow ID,
- selected variable,
- data frequency,
- latest available observation,
- forecast target period,
- date of data access,
- forecasting methods applied,
- accuracy comparison,
- selected superior method,
- final forecast result,
- interpretation,
- limitations,
- reproducibility instructions.

A README that only says “this project forecasts a TÜİK data set using R” is not sufficient.

20.16 Generic AI-Generated Forecasting Project

The project must be specifically adapted to the selected TÜİK time series. The following signs may indicate an unreviewed AI-generated template:

- generic variable names such as `value`, `time`, `data`, `series` without explanation,
- generic project title such as "Forecasting Project" without TÜİK data details,
- no dataflow ID,
- no selected variable name,
- no latest available observation,
- no forecast target period,
- no explanation of the data frequency,
- generic interpretation such as "the forecast shows an increasing trend" without linking it to the actual selected variable,
- using example data from an R package instead of TÜİK data,
- using a forecasting workflow unrelated to the required methods.

Generic structure is not automatically unacceptable, but it must be clearly connected to the selected TÜİK data and forecasting objective.

20.17 Suspiciously Polished One-Commit Submissions

A single commit is not automatically a problem. However, a one-commit repository combined with template remnants, generic content, path errors, non-working code, or inconsistent documentation will raise concerns about whether the project was generated and uploaded without real student review.

<u>Students are encouraged to build their repositories progressively and keep a meaningful commit history.</u>

20.18 Unnecessary IDE, System, or Temporary Files Left in the Repository

Repositories should be clean and professional. Files and folders such as the following should not be submitted unless there is a clear reason:

- .idea/
- .Rhistory
- .RData
- .Rproj.user/
- .DS_Store
- temporary files
- cache files
- irrelevant screenshots
- old test files
- unrelated folders

Leaving such files in the repository suggests that the submission was not checked carefully.

20.19 Direct Copying from Public Repositories, Tutorials, AI Outputs, or Other Students

Students must not copy a project from a public GitHub repository, online tutorial, AI-generated sample repository, or another student's submission and present it as their own work. The following similarities may be checked:

- project title,
- README text,
- code comments,
- function and variable names,
- repository structure,
- selected TÜİK data set,
- selected variable,
- dataflow ID,
- forecasting workflow,
- accuracy table structure,
- result sentences,
- visualization style,
- final forecast interpretation.

If a project is found to be directly copied or only superficially modified, it may not be evaluated and may be treated as an academic integrity violation.

20.20 Use of a Method Outside the Assignment Scope

The project requires students to apply and compare the specified forecasting methods.

Using additional forecasting methods is allowed only if the required methods are also considered and properly evaluated. The following cases are unacceptable:

- using only ARIMA, Prophet, machine learning, or another advanced method while ignoring the required methods,
- replacing the required methods with automated forecasting functions without explanation,
- submitting a project based mainly on classification, clustering, regression prediction without time order, or general data analysis rather than time series forecasting,
- using an AI-suggested method that does not match the assignment requirements.

Additional methods may be included as optional extensions, but they do not replace the required methods.

20.21 Final Responsibility Statement

The use of AI tools is permitted only as a support mechanism. The final submission must be the student's own reviewed, tested, reproducible, and academically responsible work.

Projects containing unremoved AI traces, template placeholders, wrong author information, another student's repository link, inconsistent README-notebook structure, non-working code, incorrect TÜİK data access, manually created data sets, missing forecasting methods, missing actual vs forecast plots, unsupported superior method selection, or direct copying from public/common repositories may be excluded from evaluation.

21. Evaluation Eligibility Criteria

Before grading, each submitted project will first be checked for basic eligibility.

A project will **not be evaluated** if it fails to satisfy any of the following compulsory conditions:

1. The GitHub repository is not public or cannot be accessed.
2. The submitted repository does not belong to the submitting student.
3. The selected TÜİK data set and selected variable were not registered in the shared Google Sheet before the analysis.
4. Another student has already registered or used the same TÜİK data set and selected variable.
5. The project does not include a working R Markdown notebook.
6. The R Markdown notebook does not run from beginning to end.
7. The TÜİK data are not accessed directly through the `tuikr` package.
8. The project uses manually downloaded, manually entered, copied-and-pasted, manually edited, or separately created data.
9. The selected data set is not a time series.
10. The project does not use the latest available TÜİK observation at the time of analysis.
11. The project does not forecast the next period immediately following the latest available TÜİK observation.
12. The submitted work is copied from another student, public repository, online tutorial, or AI-generated sample repository with only superficial changes.
13. The repository does not contain enough files, documentation, or instructions to reproduce the analysis.
14. The notebook, README, or repository contains clear unremoved AI/template placeholders showing that the submission was not reviewed before submission.

If any of these conditions are not met, the submission will be considered **ineligible for grading**. These conditions are not grading criteria.

They are minimum requirements for the project to be accepted for evaluation.

22.Evaluation Criteria

Only projects that satisfy the eligibility criteria above will be graded according to the following rubric.

Criterion	Weight
Appropriateness of the selected TÜİK time series for forecasting	10%
Correct and well-documented use of `tuikr` for TÜİK data access	15%
Correct identification of the selected variable, time variable, data frequency, latest observation, and forecast target period	10%
Appropriate exploratory time series analysis	10%
Correct application of required forecasting methods	20%
Forecast accuracy comparison using required measures	15%
Actual vs forecast plots for candidate methods	10%
Superior method selection, justification, and final next-period forecast interpretation	10%